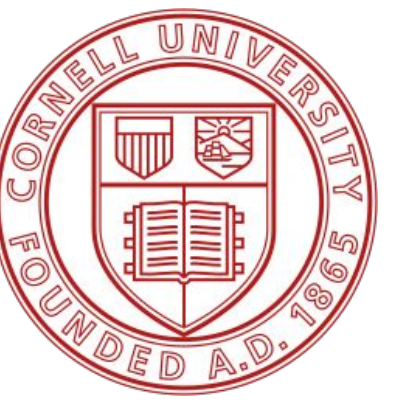


Reproducing and Extending Contrastive Decoding: From Text to Music Generation

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Introduction

- **Problem.** Decoding strategies struggle at open-ended generation
- **Motivation.** Fine-tuning LMs is computationally expensive. Inference-time strategies like **Contrastive Decoding (CD)** are an efficient way to boost generation quality

Methodology

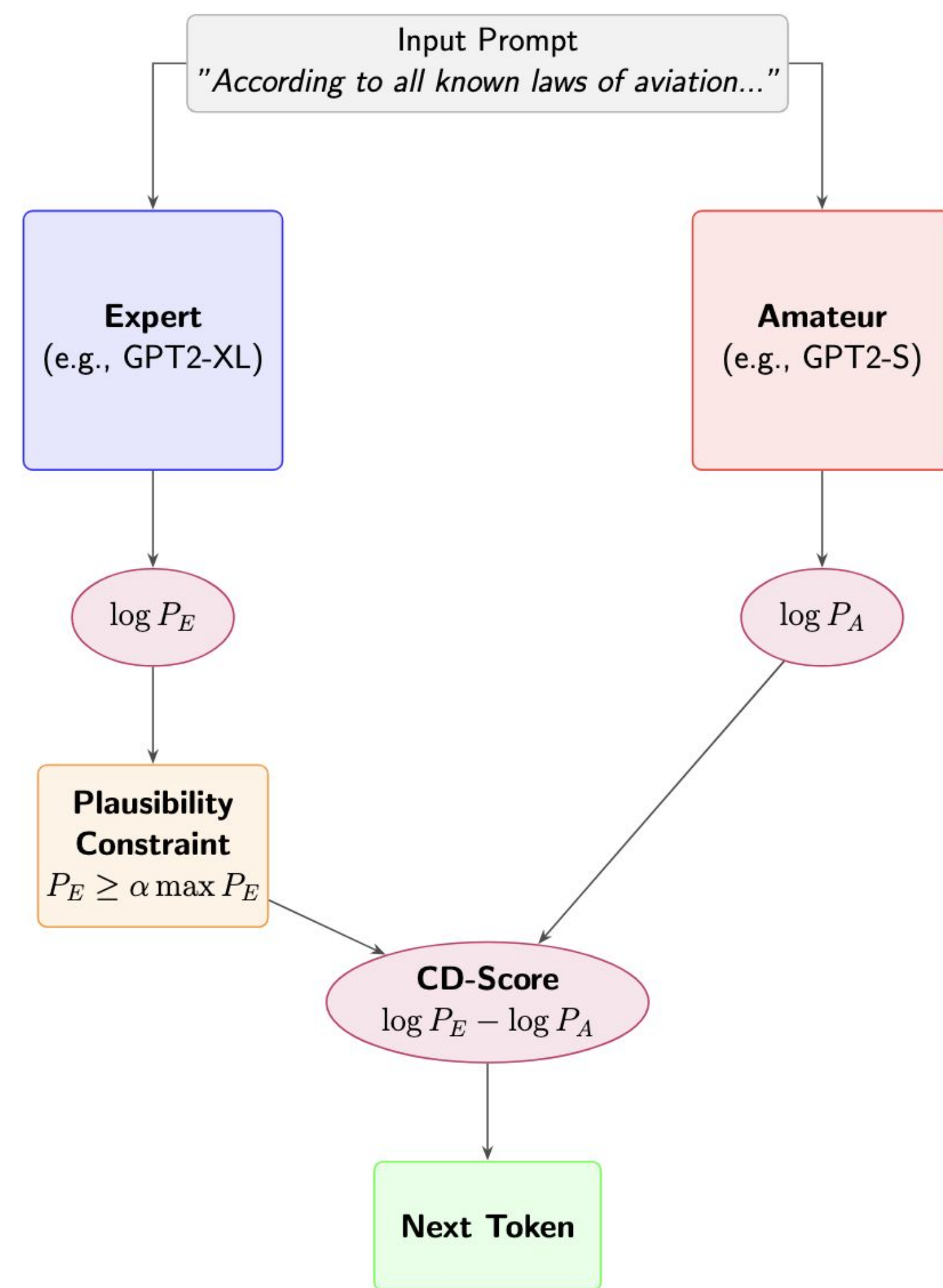


Figure 1. Flowchart for CD

Test	Expert	Amateur	Datasets
Baselines	GPT2-XL	GPT2	WikiText-103
	OPT-6.7B	OPT-125m	Wikinews
	Qwen1.5-7B	Qwen1.5-0.5B	Gutenberg
Scaling	GPT2 (S, M, L, XL)	GPT2 (S, M, L, XL)	WikiText-103
Music	MusicGen-Large	MusicGen-Small	GTZAN

Table 1. Models and datasets used for each experiment. Random samples of 1000 were taken from text datasets (999 for GTZAN)

Results

Baselines:

- CD resulted in best performance for DIV and COH for all models (Table 2)
- *Qwen1.5 (0.5B vs 7B)* to test CD performance on a newer language model

Model Scaling:

- Maximizing size gap between amateur and expert models results in near-optimal DIV and MAUVE (Figure 2)

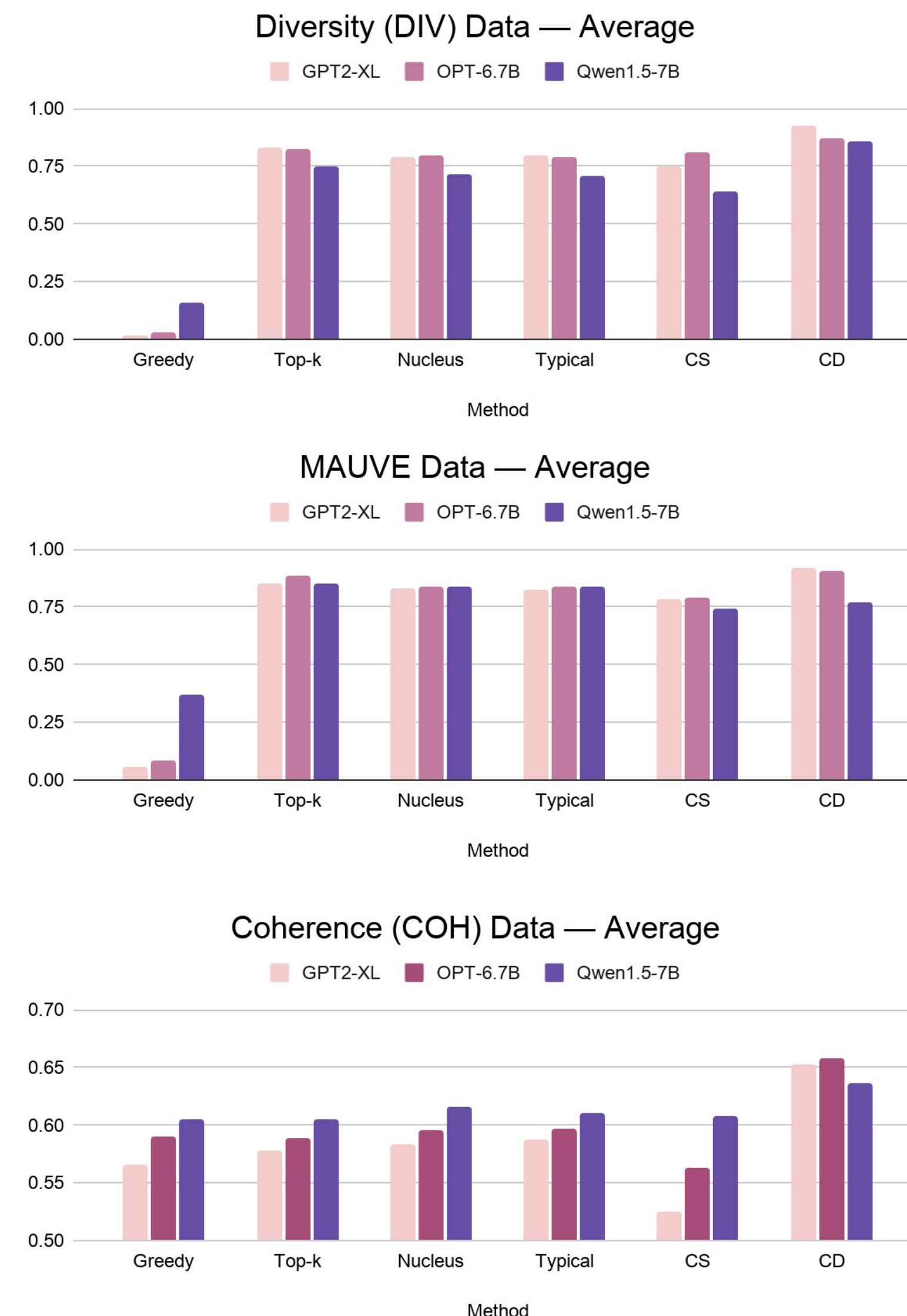


Table 2. Evaluation results for decoding methods, averaged over wikipedia, wikinews, story datasets for GPT2, OPT, Qwen1.5 families

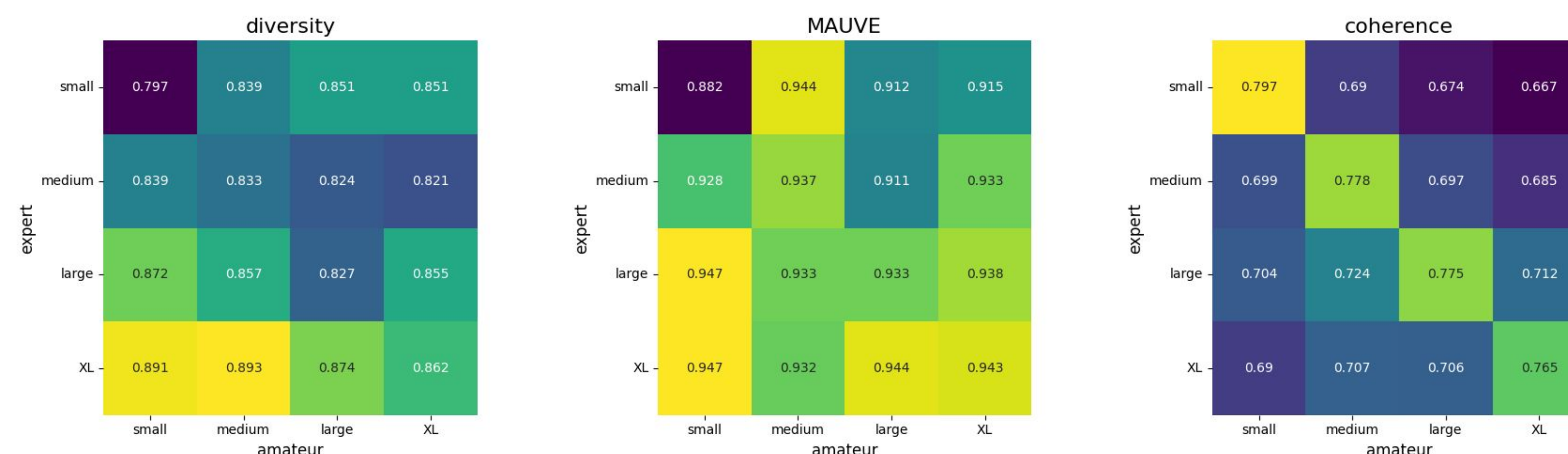


Figure 2. Generation quality using varying scales of GPT-2 for expert and amateur models. Notice the reduced variance compared to the original study.

Decoding Method	PaSST KL (↓)	FAD (↓)	MAUVE (↑)
Greedy	0.508	6.910	0.035
Top-k	0.086	1.282	0.484
Nucleus	0.114	1.940	0.328
Typical	0.113	1.874	0.321
CD, no genre	0.143	2.171	0.270
CD, same genre	0.080	1.103	0.499
CD, opposite genre	0.071	1.074	0.518

Table 3. MusicGen decoding method comparison on GTZAN. ↓ lower is better; ↑ higher is better. Best results are bolded.

Music Generation:

- CD outperforms all baseline decoding methods on GTZAN for MusicGen model (Table 3)
- Giving the amateur an opposite-genre prompt can improve results (Table 3)

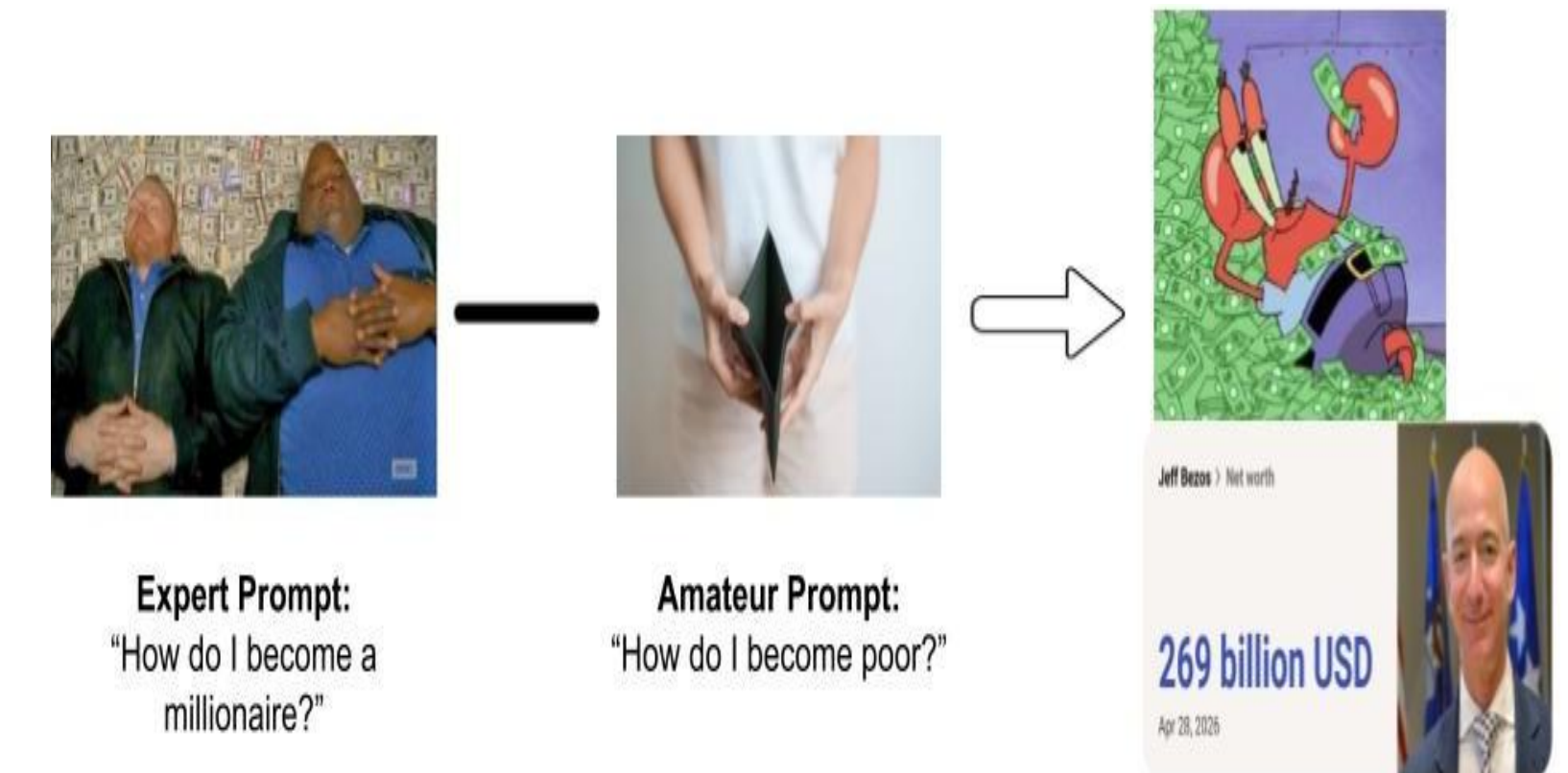


Figure 3. Opposite-prompting the amateur MusicGen model results in a stronger result (intuition).

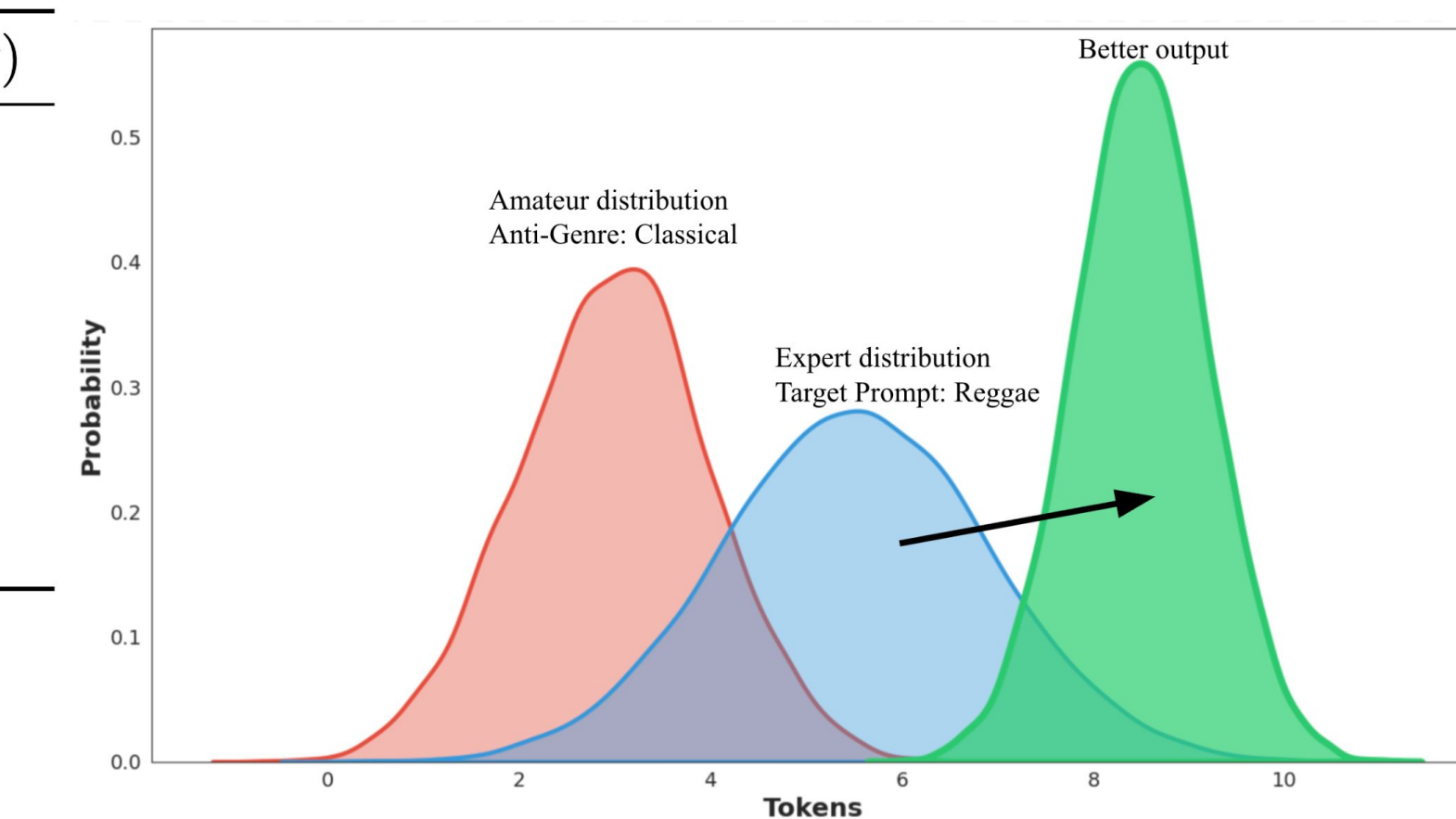


Figure 4. Subtracting amateur's "Classical" distribution from the expert's "Reggae" distribution sharpens the output to be closer to target genre

Conclusion

- CD Excels in Coherence & Diversity
- Model Scaling Matters
- CD Transfers to Music
- Anti-Genre Prompting is Effective

Citations

Li et al. 2023. Contrastive Decoding: Open-ended Text Generation as Optimization. In *Proc. ACL*.

Copet et al. 2023. Simple and Controllable Music Generation. In *Proc. NeurIPS*.